Complex Analysis Prerequisites: Definitions

Complex Numbers

Definition 1 (Complex Number)**.** *A **complex number** is an expression of the form*

$$z = x + iy, \qquad x, y \in \mathbb{R},$$

where i is the imaginary unit satisfying $i^2 = -1$. We write $x = Re(z)$ (the real part*) and $y = Im(z)$ (the* imaginary part*). The set of all complex numbers is denoted $\mathbb{C}$.*

Definition 2 (Modulus and Argument)**.** *For $z = x + iy \in \mathbb{C}$, the **modulus** (absolute value) is*

$$|z| = \sqrt{x^2 + y^2},$$

*and the **argument** $\arg(z)$ is the angle θ such that $z = |z|(\cos\theta + i\sin\theta)$, defined up to an integer multiple of 2π.*

Definition 3 (Complex Conjugate)**.** *The **conjugate** of $z = x+iy$ is $\bar{z} = x-iy$. Note that $z\bar{z} = |z|^2$.*

Definition 4 (Polar and Exponential Form)**.** *Every nonzero $z \in \mathbb{C}$ can be written as*

$$z = re^{i\theta}, \qquad r = |z|,\ \theta = \arg(z),$$

using Euler's formula (below).

Euler's Number and Euler's Formula

Definition 5 (Euler's Number)**.** *Euler's number e is defined as*

$$e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty}\frac{1}{k!} \approx 2.71828\ldots$$

[Euler's Formula] For any real number θ,

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Consequently, $e^{i\pi} + 1 = 0$ (Euler's identity).

Definition 6 (Complex Exponential)**.** *For $z = x + iy \in \mathbb{C}$, the **complex exponential** is defined by*

$$e^z = e^{x+iy} = e^x(\cos y + i\sin y).$$

Complex Functions

Definition 7 (Complex Function)**.** *A* ***complex function*** *is a map $f : D \to \mathbb{C}$, where $D \subseteq \mathbb{C}$ is the domain. Writing $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$, we identify f with a pair of real-valued functions $u, v : D \to \mathbb{R}$ (the real and imaginary parts of f).*

Definition 8 (Limit of a Complex Function)**.** *Let $f : D \to \mathbb{C}$ and let z_0 be a limit point of D. We say*

$$\lim_{z \to z_0} f(z) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |z - z_0| < \delta$ implies $|f(z) - L| < \varepsilon$.

Continuity

Definition 9 (Continuity at a Point)**.** *A function $f : D \to \mathbb{C}$ is* ***continuous*** *at $z_0 \in D$ if*

$$\lim_{z \to z_0} f(z) = f(z_0).$$

f is continuous on D if it is continuous at every point of D.

Differentiability and Holomorphy

Definition 10 (Complex Differentiability)**.** *A function $f : D \to \mathbb{C}$, with D open, is* ***complex differentiable*** *at $z_0 \in D$ if the limit*

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, where $h \in \mathbb{C}$.

Definition 11 (Holomorphic Function)**.** *A function f is* ***holomorphic*** *(or* ***analytic****) on an open set $D \subseteq \mathbb{C}$ if it is complex differentiable at every point of D.*

[Cauchy–Riemann Equations] If $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = x_0 + iy_0$, then u and v satisfy

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

at (x_0, y_0).

Convergence

Definition 12 (Convergent Sequence)**.** *A sequence (z_n) in $\mathbb{C}$* ***converges*** *to $L \in \mathbb{C}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|z_n - L| < \varepsilon$ for all $n \geq N$.*

Definition 13 (Convergent Series)**.** *A series* $\sum_{n=1}^{\infty} z_n$ ***converges*** *to* S *if the sequence of partial sums* $S_N = \sum_{n=1}^{N} z_n$ *converges to* S*.*

Definition 14 (Absolute Convergence)**.** *A series* $\sum_{n=1}^{\infty} z_n$ ***converges absolutely*** *if* $\sum_{n=1}^{\infty} |z_n|$ *converges. Absolute convergence implies convergence.*

Definition 15 (Uniform Convergence)**.** *A sequence of functions* $f_n : D \to \mathbb{C}$ ***converges uniformly*** *to* f *on* D *if for every* $\varepsilon > 0$ *there exists* N *such that*

$$\sup_{z \in D} |f_n(z) - f(z)| < \varepsilon \quad \text{for all } n \geq N.$$

Power Series and Region of Convergence

Definition 16 (Power Series)**.** *A* ***power series*** *centered at* z_0 *is an expression*

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n, \qquad a_n \in \mathbb{C}.$$

Definition 17 (Radius of Convergence)**.** *The* ***radius of convergence*** R *of a power series is given by*

$$\frac{1}{R} = \limsup_{n \to \infty} |a_n|^{1/n}.$$

The series converges absolutely for $|z - z_0| < R$ *and diverges for* $|z - z_0| > R$*.*

Analytic Continuation

Definition 18 (Analytic Continuation)**.** *Let* f_1 *be holomorphic on a domain* $D_1 \subseteq \mathbb{C}$*. If* f_2 *is holomorphic on a domain* D_2 *with* $D_1 \cap D_2 \neq \emptyset$*, and* $f_1 = f_2$ *on* $D_1 \cap D_2$*, then* f_2 *is called an* ***analytic continuation*** *of* f_1 *to* D_2*.*

[Identity Theorem] If f and g are holomorphic on a connected open set D and $f = g$ on a subset of D having a limit point in D, then $f = g$ on all of D. Consequently, an analytic continuation, when it exists, is **unique**.

Meromorphic Functions and Poles

Definition 19 (Meromorphic Function)**.** *A function* f *is* ***meromorphic*** *on a domain* D *if it is holomorphic on* D *except for a set of isolated points, at each of which* f *has a pole.*

Definition 20 (Pole)**.** *A point* z_0 *is a* ***pole*** *of* f *of order* m *if* f *can be written near* z_0 *as*

$$f(z) = \frac{g(z)}{(z - z_0)^m},$$

where g *is holomorphic near* z_0 *and* $g(z_0) \neq 0$*. A pole of order* 1 *is called a* ***simple pole****.*

The Gamma Function

Definition 21 (Gamma Function)**.** *For Re$(s) > 0$, the **Gamma function** is defined by*

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt.$$

It extends to a meromorphic function on $\mathbb{C}$ with simple poles at $s = 0, -1, -2, \ldots$, and satisfies $\Gamma(n+1) = n!$ for $n \in \mathbb{N}$.

The Riemann Zeta Function

Definition 22 (Riemann Zeta Function)**.** *For Re$(s) > 1$, the **Riemann zeta function** is defined by*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

[Euler Product Formula] For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$